Free-Energy-Driven LLM Agent for Modeling Information Exploration in Web Media

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Propose a cognitive AI agent (F-Explorer) grounded in the Free-Energy Principle (FEP) to simulate information seeking during web exploration.

1. Research Objective

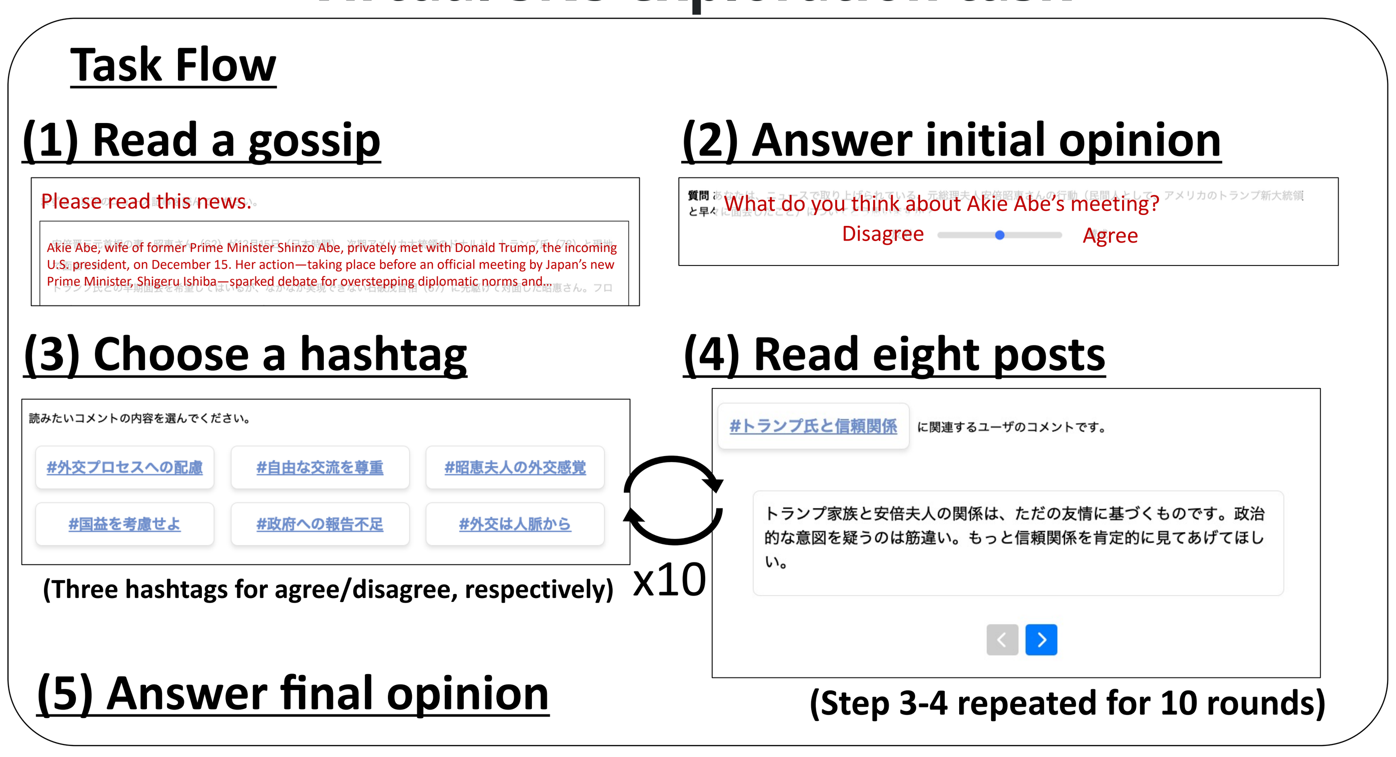
Develop a cognitive AI agent that forms and reinforces misbeliefs (e.g. pseudo-science, conspiracy) by itself through interactions with web media.

2. Confirmation Bias and SNS Task

Confirmation bias

- The tendency to selectively seek information consistent with one's existing beliefs and avoid belief-inconsistent one.
- associated with false belief formation during web exploration.

Virtual SNS exploration task



Confirmation bias in this task

- Preferentially selecting hashtags consistent with initial opinion

3. WEB-FEP

Free-energy-based model of information exploration

$$G(a) = -(V_{\text{epist}}(a) + V_{\text{prag}}(a)), \quad P(a) \propto \exp[-G(a)]$$

Epistemic value: Expected belief update in Q induced by the observation o' under action a (information gain from o')

$$V_{\text{epist}}(a) = \mathbb{E}_{Q(s',o'|a)}\left[D_{\text{KL}}\{Q(s'|o')\|Q(s')\}\right]$$

Pragmatic value: Expected alignment of observation o' with the agent's objective C

$$V_{\text{prag}}(a) = \mathbb{E}_{Q(o'|a)}[\ln P(o'|C)]$$

The balance between V_{epist} and V_{prag} represents the trade-off between information seeking and confirmatory satisfaction.

The learning rate α governing belief updating controls this balance.

Limitations

- Belief is simplified as a two-dimensional Gaussian distribution in a sentence-embedding space
- Model fit to human action sequences had not been quantitatively evaluated

[1] Fukuchi. "A Bayesian Model of Confirmatory Exploration in Text-based Web Media", Cogsci 2025.

4. F-Explorer

A cognitive LLM agent that integrates WEB-FEP with an LLM

- Uses an LLM as the agent's belief module
  - Represents linguistic meaning in a richer space

LLM Embedded in Cognitive Model (LEC)

- A major line of recent cognitive science $\times$ LLM research is data-driven, directly learning from human behavioral data
- F-Explorer embeds an LLM within an explicit cognitive model $\rightarrow$ improved interpretability

5. Formulation of F-Explorer

Texts assigned higher likelihood by the LLM are treated as more plausible under the agent's belief.

- x_i: post text; $\mathcal{X}$: fixed evaluation set, $\mathcal{X} = \{x_1, \ldots, x_N\}$
- $\ell_{\theta_t}(x)$: log-likelihood assigned to x by the LLM, $\ell_{\theta_t}(x) = \log P_{\theta_t}(x \mid p)$
- $Q_t(i)$: normalized distribution induced by likelihoods across $\mathcal{X}$

$$Q_t(i) = \frac{\exp\{\ell_{\theta_t}(x_i)\}}{\sum_j \exp\{\ell_{\theta_t}(x_j)\}}$$

- Q: distribution over $x \in \mathcal{X}$, used as a **proxy belief**

Epistemic value

$$\begin{aligned} V_{\text{epist}}(a) &= D_{\text{KL}}(Q_{t+1}^{(a)} \| Q_t) \\ &= \sum_i Q_{t+1}^{(a)}(i)\{\log Q_{t+1}^{(a)}(i) - \log Q_t(i)\} \end{aligned}$$

Pragmatic value

$$V_{\text{prag}}(a) = \mathbb{E}_{x \sim o^{(t,a)}}[\ell_{\theta_t}(x)] \approx \frac{1}{K}\sum_{j=1}^{K} \ell_{\theta_t}(x_j^{(t,a)})$$

Belief update: LoRA fine-tuning increases the likelihood of observed texts x_i.

Estimating participant-specific cognitive parameters

- Forward generative process: information environment $\rightarrow$ actions
- MLE infers latent variables from each participant's action sequence

$$\Theta^* = \arg\max_{\Theta} \log P(a_{1:T} \mid \Theta)$$

where $\Theta = \{\text{initial belief state, update strength}\}$.

Mapping Q to an opinion score

$$\text{Opinion}(Q_t) = 100 \times \frac{\sum_{i \in \mathcal{X}_{\text{pos}}} Q_t(i)}{\sum_i Q_t(i)}$$

6. Evaluation

How well can F-Explorer reproduce human exploration?

Evaluation metrics

- Log-likelihood and top-k accuracy: fit to user action sequences in the virtual SNS task
- Correlation between users' self-reported initial/final opinion scores and model-derived opinion scores

168 initial beliefs $\times$ five update-strength levels $=$ 840 configurations

7. Results

Model	$\log P(a \mid \Theta^*)$	Top-1	Top-2	Top-3	Corr. (Initial)	Corr. (Final)
(Chance level)	-17.918	.167	.333	.500	–	–
F-Explorer	-14.265	**.431**	**.668**	**.781**	**.331**	.424
WEB-FEP	-15.010	.334	.534	.711	.260	**.449**

- F-Explorer outperforms WEB-FEP in log-likelihood and top-k accuracy
- Substantial room for further improvement remains